

STANDARD OPERATING PROCEDURE

Machine Maintenance SOP

Monthly Preventive Maintenance of CNC Vertical Machining Centers

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Applies to	Greenfield Plant – Machining Cells 1–3, machines VMC-01 to VMC-06

1 Purpose

This procedure defines the monthly preventive maintenance (PM) of CNC vertical machining centers (VMCs). Its objectives are to keep the machines within their specified operating conditions, to detect wear before it causes scrap or unplanned downtime, and to ensure that maintenance work is performed safely and recorded consistently.

2 Scope

- 2.1 This SOP applies to all VMC-850 series vertical machining centers in Machining Cells 1–3 at the Greenfield Plant (asset numbers VMC-01 to VMC-06).
- 2.2 It covers the monthly PM tasks listed in Table 1 and the interval-based tasks that fall due during a monthly PM, such as the replacement of the hydraulic return filter.
- 2.3 Daily operator checks are covered by WI-2205 Daily Machine Checks – VMC. Annual geometric verification (laser calibration and ballbar testing) is performed by an external calibration provider and is outside the scope of this SOP.
- 2.4 Corrective maintenance (breakdown repair) is outside the scope of this SOP and is performed under corrective work orders in the computerized maintenance management system (CMMS).

3 References

Document	Title
SOP-00087	Lockout/Tagout (LOTO) Procedure
SOP-00056	Coolant Concentration Control
WI-2205	Daily Machine Checks – VMC
F-123-01	Monthly PM Checklist – VMC (form)
MM-VMC850	OEM Operation and Maintenance Manual, VMC-850 series, Chapter 6 Periodic Maintenance
EHS-014	Personal Protective Equipment Standard
ISO 3448	Industrial liquid lubricants – ISO viscosity classification

4 Responsibilities

Maintenance Technician: Performs the PM tasks in this SOP as an authorized employee under SOP-00087, records all readings on F-123-01 and in the CMMS, and reports any out-of-tolerance condition to the Maintenance Supervisor before the machine is released.

Maintenance Supervisor: Schedules PM work orders, ensures that technicians are trained on this SOP, reviews and signs completed checklists and approves the return to production when a nonconformity was found.

Machine Operator (affected employee): Completes the current cycle, removes the workpiece when instructed and does not attempt to operate the machine while it is locked out.

Cell Lead: Releases the machine for maintenance according to the PM schedule and confirms that the machine is ready for production after PM.

Maintenance Engineering (document owner): Maintains this SOP, reviews PM results quarterly for trends and adjusts task intervals where the data supports it.

5 Safety / PPE

WARNING Hazardous energy. The VMC contains electrical energy (400 V AC), hydraulic energy (up to 65 bar, with a charged accumulator), pneumatic energy (6 bar) and stored mechanical energy (drawbar spring pack, Z-axis counterbalance). Tasks marked "Yes" in the LOTO column of Table 1 shall only be performed after the machine has been isolated in accordance with SOP-00087.

- 5.1 The machine shall be locked out and tagged out in accordance with SOP-00087 before any guard is removed or any work is performed inside the machine enclosure or the electrical cabinet.
- 5.2 Hydraulic pressure shall be relieved and the accumulator bled down to 0 bar, as confirmed on gauge PG-1, before any hydraulic line, fitting or filter housing is opened.
- 5.3 Safety glasses with side shields and safety shoes (S3) are required for all tasks in this SOP. Cut-resistant gloves (EN 388 level C or higher) must be worn when handling tools, chips and filter elements; nitrile gloves must be worn when handling hydraulic oil, way oil or coolant.
- 5.4 Hearing protection must be worn in areas posted as hearing protection zones (above 85 dB(A)).
- 5.5 Compressed air shall not be used to blow chips out of the machine or to clean clothing or skin.
- 5.6 Oil and coolant spills must be cleaned up immediately with absorbent pads and disposed of as oily waste.
- 5.7 Work inside the machine enclosure should be performed by one person at a time; a second person may assist from outside the enclosure.
- 5.8 Only electrically qualified persons may open the electrical cabinet.

6 Tools & Materials

Item	Specification
Hydraulic oil	ISO VG 46 anti-wear hydraulic oil (HLP 46), stock no. LUB-046
Way lubricant	ISO VG 68 slideway oil, stock no. LUB-068
Hydraulic return filter element	10 µm nominal, OEM part no. HF-850-10
Torque wrench	20–100 N·m, within calibration date
Drawbar force gauge	0–30 kN with BT40 adapter, within calibration date
Belt tension meter	Frequency (sonic) type, 10–600 Hz
Dial test indicator	0.001 mm resolution, with magnetic base
Test mandrel	BT40, 300 mm, ground
Refractometer	Handheld, 0–18 % Brix, used as described in SOP-00056
Consumables	Lint-free cloths, absorbent pads, oily-waste container
Lockout devices	Personal lock, tag and hasp in accordance with SOP-00087

7 Procedure

Table 1 lists the tasks performed during the monthly PM. Record every reading on F-123-01 at the time it is taken; do not record values from memory.

Table 1 – Monthly PM tasks and intervals

No.	Task	Interval	LOTO	Step
1	Hydraulic pressure and oil temperature	Monthly	No	7.2.1, 7.2.2
2	Hydraulic oil level	Monthly	No	7.2.3
3	Way-lube tank level and lube cycle	Monthly	No	7.2.4
4	Coolant concentration	Monthly (weekly per SOP-00056)	No	7.2.5
5	Spindle drawbar force	Monthly	No	7.2.6
6	Spindle runout	Monthly	No	7.2.7
7	Pneumatic supply and water separator	Monthly	No	7.2.8
8	Hydraulic return filter replacement	Every 500 operating hours	Yes	7.4.2
9	Spindle drive belt tension	Quarterly	Yes	7.4.3
10	Way covers and wipers	Monthly	Yes	7.4.4
11	Coolant tank screens	Monthly	Yes	7.4.5
12	Guards and guard fasteners	Monthly	Yes	7.4.7
13	Hydraulic oil change	4,000 operating hours or 12 months	Yes	MM-VMC850

7.1 Preparation

- 7.1.1** Obtain the PM work order from the CMMS and a blank Monthly PM Checklist F-123-01. Enter the asset number, the date and the work order number.
- 7.1.2** Agree the maintenance start time with the Cell Lead. The operator shall complete the current cycle, remove the workpiece and return the machine to its home position.
- 7.1.3** Record the spindle run hours and the hydraulic pump operating hours from the maintenance screen of the CNC control on F-123-01.
- 7.1.4** Confirm that the torque wrench and the drawbar force gauge carry a valid calibration label. Instruments with an expired calibration must not be used.

7.2 Running checks (before isolation)

NOTE The checks in 7.2 are performed with the machine powered and the hydraulic unit running. Keep the doors closed, except where a step requires access to the spindle in setup mode.

- 7.2.1** Hydraulic pressure: With the hydraulic unit running for at least 30 minutes, read the system pressure on gauge PG-1 at the hydraulic power unit. The pressure shall be 55–65 bar. If the pressure is outside this range, adjust the pressure-reducing valve in accordance with MM-VMC850, section 6.4. If the pressure cannot be brought within range, the machine shall not be returned to production and a corrective work order shall be raised.
- 7.2.2** Hydraulic oil temperature: Read the oil temperature on the tank thermometer. The temperature shall be 40–60 °C. A temperature above 60 °C indicates a fault in the oil cooler or its fan and must be reported to the Maintenance Supervisor.
- 7.2.3** Hydraulic oil level: The oil level shall be between the MIN and MAX marks of the sight glass. Top up only with ISO VG 46 hydraulic oil (LUB-046). Oils of different grades or brands must not be mixed.
- 7.2.4** Way lubrication: Check the way-lube tank level. The level shall be above the 1/3 mark (at least 1.0 L in the 3.0 L tank). Top up with ISO VG 68 slideway oil (LUB-068). Trigger a manual lubrication cycle from the control and verify that oil reaches the X, Y and Z axis distribution blocks.

- 7.2.5** Coolant concentration: Take a coolant sample and measure the concentration with the refractometer in accordance with SOP-00056. The concentration shall be 6–8 %. If it is outside this range, correct it as described in SOP-00056 and record the corrected value.
- 7.2.6** Spindle drawbar force: Select setup mode, open the door and clamp the drawbar force gauge in the spindle using the tool release button. Take three readings and record the lowest. The clamping force shall be at least 12 kN (nominal 14 kN). If any reading is below 12 kN, the spindle shall be tagged "Do Not Operate" and the Maintenance Supervisor informed; the drawbar spring pack must be replaced before the machine is used.
- 7.2.7** Spindle runout: Clamp the test mandrel in the spindle and press the emergency stop. With the dial test indicator on the machine table, verify the spindle runout at the spindle nose and at 300 mm from the spindle nose while rotating the spindle slowly by hand. Record both readings on F-123-01.
- 7.2.8** Pneumatic supply: The air pressure at the machine regulator shall be 5.5–6.5 bar. Drain the water separator and check that the filter bowl is free of oil and water.

7.3 Isolation

- 7.3.1** Move the axes to the maintenance position (Z axis fully up, table centered), stop the spindle and switch off the hydraulic unit.
- 7.3.2** Isolate and lock out all energy sources in accordance with SOP-00087 and the energy control procedure posted on the machine (main disconnect Q1, pneumatic supply valve V-1).
- 7.3.3** Open the accumulator bleed valve and verify that gauge PG-1 reads 0 bar.
- 7.3.4** Verify zero energy by attempting to start the machine from the control panel (try-out) as required by SOP-00087. Return all controls to the off position.

7.4 Maintenance with the machine isolated

- 7.4.1** Remove the rear access guard and the hydraulic unit cover. Place the fasteners in a parts tray.
- 7.4.2** Hydraulic return filter: The hydraulic return filter element shall be replaced every 500 operating hours of the hydraulic pump, or earlier if the clogging indicator shows red. Compare the pump hours recorded in 7.1.3 with the last replacement recorded in the CMMS. Lubricate the seal of the new element with clean hydraulic oil and record the pump hours at replacement.
- 7.4.3** Spindle drive belt (quarterly): Measure the belt tension at mid-span with the belt tension meter. The belt frequency shall be 88–96 Hz. If adjustment is required, tension the belt in accordance with WI-2210 Spindle Belt Tensioning and repeat the measurement.
- 7.4.4** Way covers and wipers: Inspect the telescopic way covers and the wipers of all axes for damage, chips and coolant ingress. Replace damaged wipers; report damaged way covers to the Maintenance Supervisor.
- 7.4.5** Coolant tank: Remove chips from the coolant tank screens and the chip conveyor tray. Check the coolant level and top up with premix in accordance with SOP-00056.
- 7.4.6** Filters: Inspect filters regularly, including the coolant, hydraulic and electrical cabinet air filters, and replace them when they are dirty.
- 7.4.7** Guards: Refit all guards and covers. Tighten the M10 guard fasteners to 45 N·m with the calibrated torque wrench. Inspect the polycarbonate viewing window of the door for cracks, crazing or clouding; a damaged window shall be replaced before the machine is released.
- 7.4.8** Electrical cabinet (electrically qualified persons only): Check the door seals for damage and confirm that no coolant or chips have entered the cabinet.

7.5 Return to service

- 7.5.1** Remove all tools and materials from the machine. Account for every tool used before the doors are closed.
- 7.5.2** Remove the locks and tags in accordance with SOP-00087 and inform the operator and the Cell Lead that the machine is being restarted.
- 7.5.3** Start the hydraulic unit and run the spindle warm-up program O9000 (10 minutes). Verify that no alarms are active and that the electrical cabinet cooling unit is running.

- 7.5.4** After 15 minutes of operation, verify again that the hydraulic pressure is 55–65 bar and that there are no leaks at the filter housing or at any fitting that was disturbed.
- 7.5.5** Complete F-123-01, close the work order in the CMMS and hand the checklist to the Maintenance Supervisor for review.

8 Inspection & Acceptance Criteria

- 8.1** The machine may be returned to production only when all criteria in Table 2 are met, or when the Maintenance Supervisor has approved a documented deviation in the CMMS.

Table 2 – Acceptance criteria

Parameter	Acceptance criterion	If not met
Hydraulic system pressure	55–65 bar at PG-1	Adjust per MM-VMC850 6.4
Hydraulic oil temperature	40–60 °C after 30 min	Report; check oil cooler
Hydraulic oil level	Between MIN and MAX	Top up with ISO VG 46
Way-lube tank level	Above 1/3 mark (min. 1.0 L)	Top up with ISO VG 68
Coolant concentration	6–8 %	Correct per SOP-00056
Drawbar clamping force	Min. 12 kN (nominal 14 kN)	Tag out; replace spring pack
Spindle belt tension	88–96 Hz	Adjust per WI-2210
Pneumatic supply	5.5–6.5 bar	Adjust regulator
Guard fasteners	45 N·m	Re-torque
Leaks after restart	None visible after 15 min	Repair before release

- 8.2** Every out-of-tolerance reading must be recorded as a nonconformity in the CMMS with the measured value, the corrective action taken and the value after correction.
- 8.3** The Maintenance Supervisor shall review and sign the completed checklist within 2 working days.

9 Records

- 9.1** This SOP generates the following records: the completed Monthly PM Checklist F-123-01 signed by the technician and the Maintenance Supervisor, the PM work order in the CMMS including all readings, and the filter replacement entries with the pump operating hours.
- 9.2** Records shall be retained for 3 years from the date of the PM. Paper checklists are scanned into the CMMS within 5 working days; the paper original may then be discarded.
- 9.3** Nonconformity and deviation records are retained with the work order.

10 Revision History

Rev.	Date	Author	Description of change
A.1	2025-10-06	J. Alvarez	Initial draft for review.
A.2	2026-01-19	J. Alvarez	Review comments incorporated; drawbar force check added.
A.3	2026-03-14	J. Alvarez	Hydraulic pressure range changed to 55–65 bar; filter replacement interval set to 500 operating hours. Released.